

Composite Multi-Objective Bayesian Optimization: Benchmark Suite, Structure, and Results

tau315/composite-mobo

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1 Overview

This suite evaluates *composite* multi-objective Bayesian optimization (MOBO): instead of fitting a surrogate directly to the objective vector $f(x) \in \mathbb{R}^M$, the surrogate models an intermediate *raw response* $g(x) \in \mathbb{R}^K$ that a real simulator or solver computes, and a *known*, closed-form reduction L recovers the objectives,

$$x \longrightarrow g(x) \longrightarrow f(x) = L(g(x)).$$

The composite variant of each solver fits K per-component surrogates to g and applies L analytically; the direct variant fits M surrogates to f . Every benchmark below is a controlled A/B: identical objective, Pareto front, and evaluation budget; only the surrogate target differs.

Solvers. Each benchmark is run through up to four solver families, each in a direct and a composite variant: `standard_mobo` (qEHVI), `chebyshev_bo` (augmented-Chebyshev scalarization), `spherical_chebyshev_bo` (a spherical scalarization for high dimension), and `batched_morbo` (trust-region MORBO).

Suite gating. “Low”-suite benchmarks ($\lesssim 10D$) run `standard_mobo/chebyshev_bo/batched_morbo`; “high”-suite benchmarks run `spherical_chebyshev_bo/batched_morbo` (plain-kernel scalarization is a weak test at high dimension). Reported numbers are the composite-minus-direct percentage change in final dominated hypervolume, paired per seed, at n seeds; * marks $p < 0.05$ (paired t -test).

2 Benchmarks and their composite structure

2.1 Synthetic composite functions

These have a decomposable objective by construction and serve as mechanism probes.

- **DTLZ2** (6D/2obj, 100D/2obj, 600D/2obj; also a 6D/5obj variant). Objective $f_m = (1 + g) \prod \cos(\cdot)$ with $g = \sum_j (x_j - 0.5)^2$; the raw response exposes the per-variable distance terms, so g is a sum of one-dimensional components. The canonical decoupled case.
- **Ackley–Griewank** (6D, 50D / 2obj), **Five-Ackley** (6D/5obj), **Langermann–Ackley** (6D/2obj), **Projected Langermann** (500D/2obj): additive synthetic multi-objective landscapes whose raw response exposes the additive terms feeding each objective.

2.2 Real-world benchmarks

- **Summit SnAr reactor** (4D/2obj). Nucleophilic aromatic substitution in a plug-flow reactor (Felton et al. 2021; Hone et al. 2017 kinetics). Raw response: the five outlet species concentrations plus the product molar flow; reduction: space-time yield (linear in the product flow) and E-factor (a nonlinear *ratio* of species masses).
- **Nanoparticle RGB** (6D/3obj). Mie-scattering multilayer nanoparticle designed for RGB-selective structural color; raw response is the per-wavelength scattering response reduced to three color objectives.
- **CORT TG119 radiotherapy** (418D/3obj). Intensity-modulated radiotherapy dose optimization on the TG119 phantom; the raw response is the per-voxel dose, reduced to target-coverage / organ-sparing objectives.
- **Penicillin fermentation** (7D/3obj). Fed-batch bioreactor; raw response is a checkpointed fermentation trajectory reduced to yield, CO₂, and time.

2.3 Power-flow benchmarks (MOBO-HighDim contribution)

Real IEEE 14-bus optimal power flow from the CEC2021 RWCMOP suite. The raw response is the per-bus power injection vector, computed by a single admittance solve $I = YV$, $S = V \odot \bar{I}$; because the admittance matrix is sparse, each per-bus quantity depends only on its local neighbors, and the objectives are decoupled sums over them.

- **RCM40** (34D/2obj): active/reactive power loss.
- **RCM46** (34D/4obj): fuel cost, active loss, reactive loss, voltage deviation.
- **MOOPF-5** (34D/5obj): RCM46’s four objectives plus the Kessel–Glavitsch L-index voltage-stability margin. A *constructed* benchmark (not a CEC2021 case, which max out at RCM46’s four objectives), built to test whether more decoupled objectives amplify the composite advantage.

2.4 Controlled decoupling benchmark

Blocked DTLZ2 (`benchmark_dtlz2_blocked_reference.py`). DTLZ2 with the objective, input dimension, and objective count held fixed while the distance term g is partitioned into $K = \lceil (d-1)/b \rceil$ contiguous blocks; each raw component is a partial sum over b inputs. Because $\sum_i \text{block}_i = g$ for every b , the objective is identical across the sweep – only the effective input-dimension of the raw components (b/d) varies, isolating decoupling from all other factors.

3 Results

Benchmark (best composite pair)	Dim/Obj	n	Composite ΔHV
<i>Power flow (MOBO-HighDim)</i>			
MOOPF-5 (batched_morbo)	34/5	20	+23.5%*
(spherical_chebyshev)	34/5	13	+14.6%*
RCM46 (spherical_chebyshev)	34/4	18	+19.0%*

Benchmark (best composite pair)	Dim/Obj	n	Composite Δ HV
(batched_morbo)	34/4	20	+5.4%*
RCM40 (batched_morbo)	34/2	20	+0.8%*
<i>Synthetic</i>			
DTLZ2 (batched_morbo)	6/2	20	+5.3%*
Langermann–Ackley (standard_mobo)	6/2	20	+8.0%*
Five-Ackley (batched_morbo)	6/5	20	+5.0%*
DTLZ2 (batched_morbo)	100/2	20	+0.3%*
DTLZ2 (5obj) (batched_morbo)	6/5	20	+0.5%*
DTLZ2 (batched_morbo)	600/2	20	+0.1%
Ackley–Griewank (standard_mobo)	6/2	20	+2.7%*
Ackley–Griewank (batched_morbo)	50/2	20	+0.1%
Projected Langermann (spherical)	500/2	8	+0.5%*
<i>Real-world (chemistry / physics / medical)</i>			
Nanoparticle RGB (standard_mobo)	6/3	20	+0.2%
Summit SnAr (batched_morbo)	4/2	20	+0.3%*
Penicillin (batched_morbo)	7/3	20	−2.5%*
CORT TG119 (spherical_chebyshev)	418/3	20	+1.5%*
(batched_morbo)	418/3	20	−2.0%*

Table 1: Composite-minus-direct final-hypervolume change, best/most-complete solver pair per benchmark. *: $p < 0.05$, paired t -test.

4 Key findings

1. **The largest wins are on decoupled power-flow problems, and scale with objective count.** On the identical IEEE 14-bus system the composite advantage grows RCM40 (+0.8%, 2 obj) $\rightarrow$ RCM46 (+5.4%, 4 obj) $\rightarrow$ MOOPF-5 (+23.5%, 5 obj). Each added decoupled objective handicaps direct modeling (more coupled surrogates) more than composite.
2. **Method dependence is real.** CORT TG119 flips sign by solver: composite *loses* −2.0% on `batched_morbo` but *wins* +1.5% on `spherical_chebyshev.bo`. The composite advantage is strongest with trust-region acquisition and weakest with plain scalarization.
3. **Chemistry / trajectory benchmarks do not benefit.** SnAr ($\approx$ tie), Penicillin (−2.5%), and CORT-on-morbo are cases where the raw response is one globally-coupled state (a reaction/fermentation trajectory, a global dose solve); the intermediates are no easier to model than the objectives.
4. **Decoupling, not reduction nonlinearity, is the discriminator.** RCM46 wins with a purely *linear* reduction (sums), while SnAr’s *nonlinear* E-factor ratio still ties. Across the suite, the effective input-dimension of the raw response separates wins from losses.
5. **Causal confirmation (blocked DTLZ2).** Holding the objective, dimension, and objective count fixed and varying only the raw-response block size, the composite advantage falls monotonically from +11.9% (fully decoupled) to +7.4% (fully coupled), Pearson $r = -0.997$

(Figure in `ablation_results/_decoupling_sweep/`). Decoupling *causes* the advantage; it is not merely correlated with it.